

Week 9 Social dilemmas

Mon 4 May 26 - Sun 10 May 26

This week

Overview

Often there is a discrepancy between what is good from the perspective of an individual and what is good from the perspective of the collective (group/society). We discuss how game theory helps us to understand and analyse such scenarios. We will also tie up some loose ends in game theory, including the handling of sequential games and how we can marry game theory and reinforcement learning for two player zero sum games.

Your Learning Journey

Own-time



Real-time



1 TASK



Wrap-up



Own-time

Your schedule for the week

Tasks	Complete By/Before	Time Expected
Listen to the mini podcast [experimental]	Before the seminar	15 minutes
Participate in the seminar & tutorial	End of the week	2 hours + 2 hours
Complete the essential tutorial components if you didn't manage this during the tutorial	End of the week	[individual]
Consolidate your learning with the extended podcast and chat space [experimental]	End of the Week	1 hour
Start to revise for the final exam	End of the Week	[individual]

Mini-Podcast: MiniMax-Q M4A

AI-generated mini podcast setting the scene for evolutionary game theory (1'50'). While this is originally a concept of evolutionary biology we will reinterpret it in the context of learning.

The AI-generated podcasts are generated from resources curated by me and vetted before release. However, they are still not fully authoritative resources and you should not blindly trust them. In fact, part of our experiment with the podcasts is that you are meant to come back with questions about what is said in the podcasts that we can then discuss in the seminar and in the tutorials. In particular, for these discussions you should try to identify elements that you think might be incorrect!

3.0 MB

 [Mini-Podcast: Extensive Form](#) M4A

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2.7 MB

Real-time



Mark as done

Seminar Resources

Most of the material for this section was bundled with the resources for last week's seminar. Below are three additional "mini topics" that we will discuss. We will barely be able to scratch the surface of these topics in the time available but you should at least be aware that they exist!

 [MinMax Q](#) PDF

1.5 MB

 [Extensive Form](#) PDF

688.1 KB

Tutorial Resources

 [Lab Week 9 Instructions](#) PDF

44.1 KB

 [Week 9 Lab Zoom Link for Suzhou](#)

Wrap-up



 [AI Review Podcast \[experimental \]: Extensive Form](#) M4A

9.5 MB

 [AI Review Podcast \[experimental \]: MiniMax-Q](#) M4A

WARNING This podcast is probably not particularly useful since it does not correctly focus on the essential aspects of MiniMax-Q. It also contains at least one mistake; claiming that a zero sum game requires a mixed strategy. This is plainly incorrect.

You know that these podcasts are AI-produced from curated resources. In this particular instance, I have been unable to prompt the AI successfully into generating a useful podcast despite many attempts with increasingly specific instructions. (I'm giving up at this point, please refer to the seminar recording instead :) The chat space should still be useful.

7.5 MB



AI Chat Space [experimental]: MiniMax-Q

Please keep in mind that these experimental AI chat spaces are not authoritative sources. While they work from curated authoritative sources they can and will make mistakes. Use them for guidance but make sure that you always check everything using the authoritative literature sources.



AI Chat Space [experimental]: Extensive Form

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